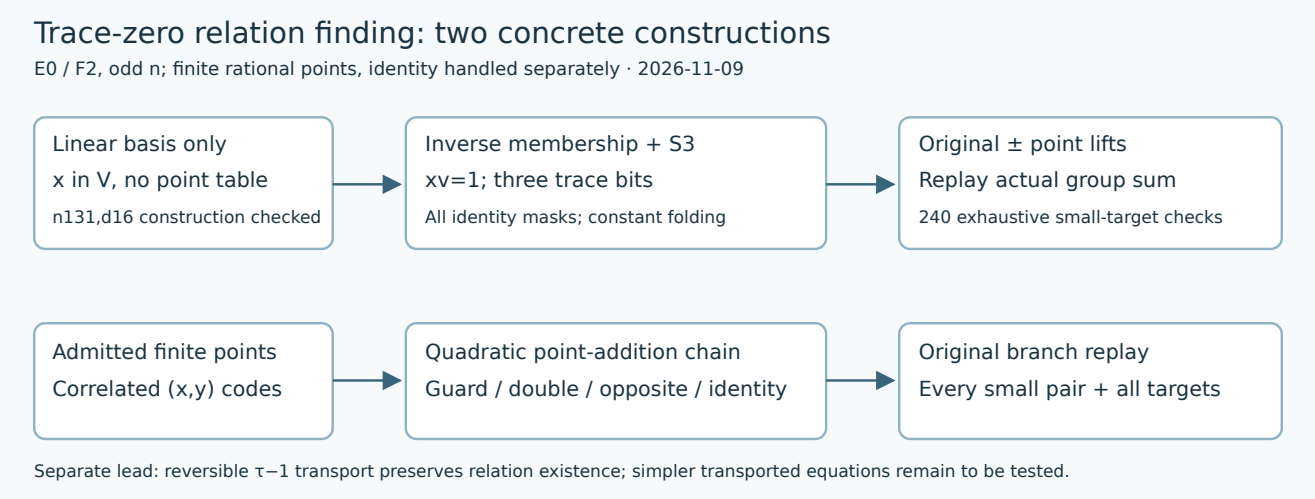
Trace-zero relation models and tau transport

2026-11-09 · isogeny-labs/crypto · PR3744 · unsigned local evidence

Implemented: a basis-only decomposition oracle without a finite factor-base table; a complete finite-base full-point chain; inverse/half-trace membership; opt-in constant specialization; reversible tau-minus-one transport. Twenty focused checks, ten original regressions and seven prior-lead regressions pass. The production library type-check passes. These are correctness checks and static constructions. No comparative experiment panel ran.



Exact working scope

E0: $y^2+xy=x^3+1$ over F2; odd n. The full trace-zero object in the Weil restriction has dimension n−1. Its rational point group T_n equals4E0. Models here describe these finite rational points, with a separate identity branch. They are not general scheme charts over arbitrary extension algebras. ECC2K-130 uses n131, the original repository field, G and public Q.

The new subspace oracle takes a linear basis, target and arity. It defines B={affine P in T_n : x(P) in V}, without enumerating B. Free coefficients and inverse membership constrain each summand. All intermediate-identity masks are kept. Partial sums are not restricted to V. Every SAT tuple is independently lifted to both original signs and replayed in the group; all signs must fail before that tuple is blocked. Complete exhaustion refutes; caps or unsupported inputs remain inconclusive.

Every target of the n5/7 kernels, d0/2/3 and arities2/3 agrees with separate exhaustive enumeration:240 calls, no point table passed to the solver. Wide d16,m3,all-affine constructions give:

n	SAT variables	S3 AND gates	CNF clauses	XOR rows
83	10794	5485	28666	1331
131	16890	8509	44938	2099

Membership gates are included in total variables/clauses, excluded from the S3 AND column. One static construction per row; no runtime or scaling claim.

Inverse witnesses: exact x-image, degree two

For $x \neq 0$ put $v = 1/x$ and $H(a) = \Sigma\, a^{(2^{(2i)})}$, $i = 0,\ldots,(n-1)/2$. Impose $xv = 1$, $\mathrm{Tr}(x) = 0$, $\mathrm{Tr}(v) = 0$ and $\mathrm{Tr}(xH(v^2)) = 0$. This reuses known validation algebra: US7257709 condition iii already supplies the last test for $b = 1$ on partially valid cofactor-four keys. No mathematical novelty is claimed.

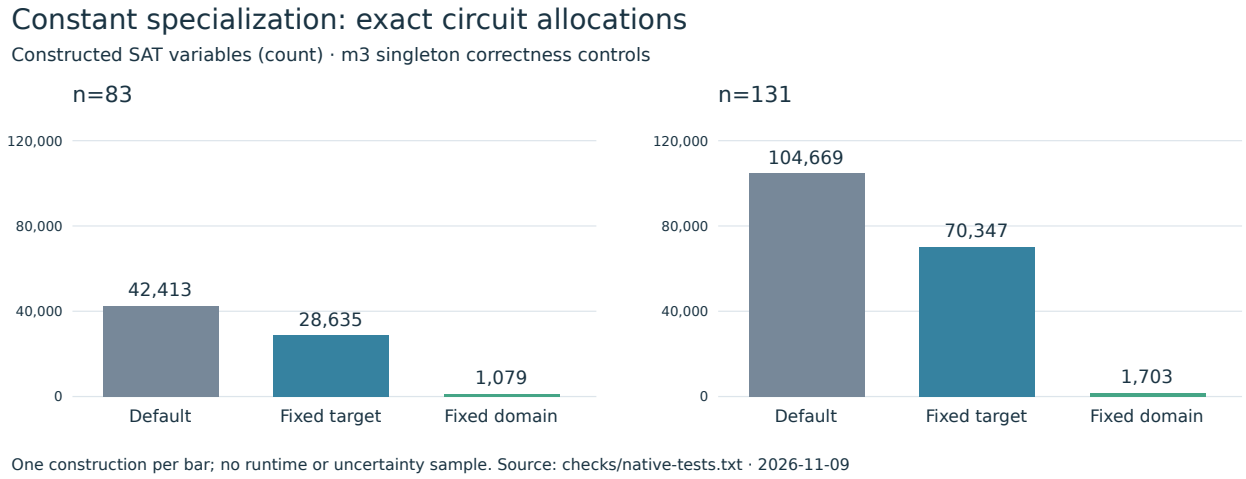
Rationality is $\mathrm{Tr}(x+v^2) = \mathrm{Tr}(x) + \mathrm{Tr}(v) = 0$. With $s = \sqrt{x}$ and $u = sH(v)$, $u^2 + su + 1 = 0$ and $u^2 + u^{-2} = x$. Thus u is a half-point abscissa; its trace is $\mathrm{Tr}(xH(v^2))$. Vanishing is the second halvability condition. Original point lifts and full group trace are independently replayed.

The n3 counterexample $x = z$, $v = z^2 + 1$ passes $\mathrm{Tr}(x)$ and the final trace but fails $\mathrm{Tr}(v)$. Dropping rationality admits a false x. This finite image of T_n under x is not all rational points of a geometric Kummer quotient. Exhaustive x-image sizes at n3/5/7/9 are0/5/14/63. Raw/compact assignments have one inverse per x; original point lifts have two signs.

n131 model	Variables	Rows	Degree	AND aux
Inverse compact global	260	132	2	16900
Point compact global[1]	260	131	3	43517
Inverse compact d16	146	132	2	2080
Point compact d16[1]	146	131	2	2200

[1] Historical counts in previous studies, unchanged. Matching x-domains project to the same x set, with different witnesses. Inverse d16 also has an extra row and more ANF terms (3627 versus3076). Counts do not determine solving cost. Both wide inverse-membership singleton controls return **UNKNOWN** at2seconds/2000conflicts: n83 has21331variables, n131 has52531.

Compile constants before multiplication



Target-only and fixed-domain modes preserve all small projected S3 solutions. Only fully fixed-domain controls were solved here: both SAT with field replay, zero multiplication gates. They are x-only systems without inverse membership and do not resolve the UNKNOWN controls above. One construction per bar; exact recorded counts, no statistical interval or timing ratio.

Complete point chains and a Z[tau] presentation lead

The finite-base point chain retains correlated (x,y) codes. Generic addition has a nonzero denominator guard. Doubling, opposite points and identity-plus- point use distinct exact branches. Each model replays every branch and the original group sum. All100 ordered affine n5 pairs plus wrong-output controls, all1000 triple branch paths, every small target at arities2/3/4 and sign- restricted bases agree. Pinned n83/131 identity paths use830/1310variables and replay G+(−G)+G=G. This completes the finite-base branch core of TZL-03; unenumerated full-point membership is not yet attached to it.

For E0/F2, $\tau^2+\tau+2=0$. Existing $\delta=\tau-1$ has norm4, and $(t-1)(-t-2)+(t^2+t+2)=4$. New inverse_cofactor_image applies $[4^{-1}](-\tau-2)$ on the odd-order kernel, rejects nonkernel points and handles the identity. Complete n5/7 checks establish bijection and the same three- term relation existence on a signed base and its transported image. Wide checks replay n131 G/publicQ and n83 G/17G in both directions, including torsion-coset rejection.

Transporting both base and target preserves the number of relations. The candidate beyond orbit compression is a different equation presentation: Frobenius is linear in field coordinates, while a lifted δ map introduces guarded addition. Does a direct base or its δ-preimage give cheaper complete solving? The transported x set is not assumed linear. A charged comparison must retain equal base sizes, corresponding targets and every failed attempt.

Evidence, certificates and execution dependency

README.md contains exact fixture identities, formulas, source links and reproduction commands. checks/ retains20 focused tests,10 original and7 lead regressions, root check, two corrected compile attempts and all UNKNOWN. SOURCES.sha256 pins current code; earlier study evidence remains historical. counts.json and plot_counts.py reproduce the graph without new computation.

Evaluated certificate IDC1h2a629cbcbb26bb0c records the degree4 integral norm expansion behind the inverse lift; IDC1he63d40398a440139 records the degree2 tau factorization. Both use32 issued plus32 local replay points over$2^{61}-1$. They certify the expansions only, not the Frobenius/halving theorems, characteristic-two subgroup proof or cost. No signed independent standing or formal proof is attached.

Flow: all listed local checks completed, no matched benchmark rows attempted. The264-row panel is prepared at source3ae8b2124 with its original executable hash15c4d1b193999153c6ddc6ec94959167c8f4d016be197b55b26d6993001b1126. Checkpoint upload fails on missing credentials. The explicit, capped local EKS/S3 exception is pending. No pilot or cloud job launched; new candidates need their own pre-data protocol. Whole-root and benchrec gaps on PR3744 remain open. Scoreboard, timeline and performance graphs retain prior values.

Sources: Kosters–Yeo1503.08001; primary US7257709 condition iii (https://patents.justia.com/patent/7257709); production BinaryCurve halving/ addition, previous trace-zero studies; standard Frobenius characteristic equation (Sutherland18.783 Lecture13, reference not newly fetched here). KB unavailable, wrapper127. Existing atlas families cover these mechanisms; no global novelty assertion. Next deciding checks compare complete successful work on identical bases for implicit membership, point chains and τ transport.